

Draft Literature Review: The Impact of Model Compression on Demographic Fairness in Face Verification

Overview

Demographic fairness in face recognition has become a major research concern because modern systems can achieve high aggregate accuracy while still producing systematically different error rates across demographic groups.[cite:34] In verification settings, these differences are most meaningfully expressed through threshold-dependent metrics such as False Match Rate (FMR) and False Non-Match Rate (FNMR), which reveal subgroup disparities that overall accuracy can obscure.[cite:34] NIST evaluation programmes have been especially influential in making demographic differentials a standard part of face recognition assessment rather than a secondary ethical consideration.[cite:4]

This dissertation is positioned at the intersection of three strands of work: demographic auditing in face verification, efficient model compression for deployment, and image-quality-aware biometric evaluation.[cite:4][cite:7][cite:60] The core research question is whether structured pruning and INT8 quantisation preserve, widen, or reduce demographic disparities in face verification performance when applied to ArcFace-based backbones under the LFW restricted protocol.[cite:7][cite:42][cite:61]

Demographic fairness in face recognition

The literature consistently shows that demographic unfairness in face recognition is not attributable to a single cause.[cite:34] Instead, disparities are generally understood as emerging from the interaction of training data imbalance, image quality, sensor and capture conditions, algorithm design, and evaluation methodology.[cite:34] This is particularly important in biometric verification because systems that appear strong on headline accuracy may still exhibit materially different false match and false non-match behaviour across groups.[cite:34][cite:4]

Recent review work on demographic fairness in face recognition synthesises these concerns by showing that race, gender, age, and other soft-biometric factors remain significant axes of performance variation across many face analysis systems.[cite:34] This body of literature has pushed researchers away from reporting only aggregate benchmark scores and toward disaggregated evaluation protocols that expose subgroup performance patterns.[cite:34] For this reason, a fairness-aware face verification study now typically requires per-group reporting and a clear discussion of operating thresholds, not only a single overall verification score.[cite:34][cite:4]

Audit and benchmark work

The most influential operational work on demographic fairness in face recognition comes from NIST's FRVT and FRTE programmes.[cite:4][cite:13] These evaluations report demographic differentials in face recognition systems at scale and have helped establish the practical importance of analysing false matches and false non-matches across demographic categories.[cite:4] Their main relevance to this dissertation is methodological: they provide a strong precedent for analysing subgroup error rates rather than relying solely on aggregate performance.[cite:4]

NIST-style evaluations also show that demographic error differentials interact with conditions such as image quality and acquisition setup.[cite:13] This is important because it suggests that fairness should not be treated as a static property of a model alone, but as something shaped by the data accepted into the recognition pipeline.[cite:13][cite:34] As a result, the use of a quality-based gating method such as FaceQNet is not merely an auxiliary engineering choice; it may meaningfully affect the observed fairness profile of the system.[cite:60][cite:66]

Datasets and verification protocols

A recurring issue in the fairness literature is that LFW, although historically important, was not designed primarily for demographic fairness evaluation.[cite:34] It remains useful as a standard face verification benchmark and offers a well-known restricted 10-fold protocol, but its demographic distribution and subgroup granularity limit the strength of fairness claims that can be made from it alone.[cite:61][cite:69][cite:34] This means that LFW is suitable for establishing a controlled baseline, but any fairness conclusions drawn from it should be presented cautiously.[cite:34]

More recent fairness-focused datasets and benchmarks such as FairFace, RFW, and BFW were designed to improve subgroup coverage or make demographic comparisons more explicit.[cite:34][cite:42] Their significance lies in enabling more balanced verification analysis and in making demographic disparities easier to detect and interpret.[cite:34] Even if the dissertation remains centered on LFW for feasibility and comparability, these datasets are important in the literature review because they clarify what stronger fairness benchmarking looks like.[cite:34]

Fairness metrics and evaluation practice

Recent face verification research increasingly treats subgroup-specific error rates as the most operationally meaningful fairness indicators.[cite:34] In verification tasks, FMR and FNMR are particularly important because they describe the two principal error types that matter in biometric decision systems, and both vary with threshold selection.[cite:34][cite:4] This makes them more informative for fairness analysis than a single summary accuracy figure.[cite:34]

The FDR framework is especially relevant to this dissertation because it extends subgroup analysis beyond raw error gaps and provides an additional formalisation of fairness differences in face recognition systems.[cite:34] Together, FMR, FNMR, and FDR support a more nuanced evaluation of whether compression changes the distribution of harms across groups rather than simply lowering average accuracy.[cite:7][cite:34] This is consistent with current fairness-

oriented practice, which increasingly emphasizes disaggregated metrics, threshold-aware reporting, and transparent discussion of subgroup sample limitations.[cite:34][cite:73]

Fairness-aware architectures and mitigation

Some recent work attempts not only to measure bias, but also to reduce it through model design or training strategies.[cite:42][cite:34] One notable example is FairGRAPE, a fairness-aware pruning method for face-related classification tasks that explicitly accounts for group-wise importance during pruning and reports reduced disparity compared with conventional compression baselines.[cite:42] Although this is not a face verification study, it is highly relevant because it demonstrates that pruning decisions can affect demographic performance unevenly.[cite:42]

More broadly, the literature suggests that architecture choice itself can influence fairness outcomes.[cite:34] This provides a strong rationale for comparing ResNet-50 and MobileNetV3 in the dissertation, since a smaller or more efficiency-oriented backbone may not degrade uniformly across groups under compression.[cite:34] The architecture comparison therefore contributes not only to efficiency analysis but also to the question of whether some backbone families are more fairness-robust under deployment constraints.[cite:34]

Model compression and fairness

The most directly relevant compression-and-fairness evidence identified in the reviewed literature comes from outside the face verification task itself.[cite:7] Stoychev and Gunes study the effect of model compression on fairness in facial expression recognition and show that pruning can amplify subgroup disparities, while quantisation can have a smaller or different effect depending on the metric and setting.[cite:7] Although facial expression recognition is not equivalent to identity verification, this paper provides a strong conceptual precedent for investigating whether compression changes demographic fairness without necessarily destroying aggregate task performance.[cite:7]

FairGRAPE further strengthens this motivation by showing that pruning is not fairness-neutral in face-related deep models.[cite:42] However, the available verified literature does not clearly show an exact prior study that combines structured pruning or INT8 quantisation with demographic fairness auditing in a face verification pipeline using verification-centric metrics such as FMR, FNMR, or FDR.[cite:7][cite:42][cite:34] On that basis, the dissertation's novelty claim appears plausible, though it should be framed carefully as a gap in the verified accessible literature rather than as absolute proof that no such study exists.[cite:34][cite:7]

Compression for face recognition backbones

Compression work on face recognition systems more commonly focuses on efficiency, latency, storage reduction, or deployment on constrained hardware than on fairness.[cite:48][cite:50] These studies typically investigate pruning, low-bit quantisation, or lightweight architectures to preserve recognition performance while reducing memory use and inference cost.[cite:48][cite:50] Their relevance to this dissertation is mainly methodological, since they help justify the practical feasibility of pruning and INT8 quantisation for ArcFace-style recognition backbones.[cite:48][cite:50]

This literature also reinforces the practical value of comparing a heavier architecture such as ResNet-50 with a more efficiency-oriented model such as MobileNetV3.[cite:48][cite:50] Even where fairness is not discussed explicitly, any evidence that compression affects robustness differently across datasets or image conditions is indirectly relevant because those shifts may translate into demographic disparities when subgroup distributions differ.[cite:48][cite:50][cite:34]

Quality assessment and FaceQNet

Image quality is widely recognised in biometrics as one of the strongest determinants of recognition performance.[cite:66] The FaceQnet line of work addresses this by estimating a scalar quality score that predicts the usefulness of a facial image for recognition, and its authors report that the measure correlates strongly with downstream face recognition performance.[cite:60][cite:64][cite:66] This makes FaceQNet a logical tool for testing whether quality filtering changes both overall verification performance and subgroup fairness outcomes.[cite:60][cite:66]

Quality filtering, however, introduces an important fairness question of its own.[cite:34][cite:66] If lower-quality images are not uniformly distributed across demographic groups, then a quality gate may reduce error rates overall while disproportionately excluding images from certain subgroups, thereby changing the observed fairness profile of the verification system.[cite:34][cite:66] In this sense, FaceQNet is not only a performance-enhancement mechanism, but also an intervention whose demographic consequences require explicit auditing.[cite:60][cite:66]

Gap and dissertation contribution

Taken together, the literature supports three conclusions.[cite:34][cite:7][cite:42] First, demographic fairness auditing in face recognition is well established and strongly grounded in subgroup error analysis.[cite:4][cite:34] Second, model compression is widely studied for efficiency but rarely paired with demographic fairness evaluation in face verification.[cite:48][cite:50][cite:7] Third, image quality is a known driver of biometric performance and may interact with fairness in ways that remain underexplored in lightweight verification settings.[cite:60][cite:66][cite:34]

The dissertation therefore occupies a credible research gap: it examines whether structured pruning and INT8 quantisation alter demographic fairness in face verification, while also testing whether FaceQNet-based filtering changes that relationship.[cite:7][cite:42][cite:60] Its contribution is not the invention of a new backbone or fairness metric, but a systematic evaluation of how deployment-oriented efficiency techniques interact with subgroup verification errors in a practical, constrained setting.[cite:7][cite:34][cite:42]